

Section 2: Economic Lock-In (C2)

Assumptions Used

Assumption 1 (Documented Productivity Gains (A2)). *For an organization that has adopted AI at integration depth $d \geq 1$:*

1. *Annual net payoff (revenue minus all costs, including AI operating costs) satisfies $\Pi_{\text{AI}}(d) \geq (1 + \gamma)\Pi_0$ for some fixed $\gamma > 0$ (empirically $\gamma \in [0.3, 0.55]$).*
2. *Pre-AI net payoff satisfies $\Pi_0 > 0$ (the organization is viable without AI).*
3. *After reversion to pre-AI, net payoff returns to Π_0 (no other reversion-side benefits or costs).*

Assumption 2 (Super-Linear Switching Costs (A4)). *Removing AI at integration depth $d \geq 1$ incurs a one-time switching cost satisfying $S(d) \geq c d^2$ for some fixed $c > 0$.*

Definitions

Definition 1 (NPV of Reversion). *For an organization at depth d with continuous discount rate $\rho > 0$, the net present value of reverting to pre-AI is:*

$$V_{\text{rev}}(d, \rho) = -S(d) + \int_0^\infty e^{-\rho t} \Pi_0 dt = -S(d) + \frac{\Pi_0}{\rho}.$$

The NPV of continuing with AI is:

$$V_{\text{cont}}(d, \rho) = \int_0^\infty e^{-\rho t} \Pi_{\text{AI}}(d) dt = \frac{\Pi_{\text{AI}}(d)}{\rho}.$$

Reversion is economically rational only if $V_{\text{rev}} > V_{\text{cont}}$, i.e., $V_{\text{rev}} - V_{\text{cont}} > 0$.

Statement to Prove (C2)

Theorem 1 (Economic Lock-In). *Under Assumptions ?? and ??:*

- (i) *For any organization at integration depth $d \geq 1$ and any discount rate $\rho > 0$, the NPV differential satisfies:*

$$V_{\text{rev}}(d, \rho) - V_{\text{cont}}(d, \rho) \leq -c d^2 - \frac{\gamma \Pi_0}{\rho} < 0.$$

- (ii) *Therefore reversion is never economically rational for any organization at depth $d \geq 1$ and any $\rho > 0$: AI adoption is individually economically irreversible.*
- (iii) *As depth d increases, the economic case against reversion strengthens: the NPV differential is bounded above by $-c d^2 - \gamma \Pi_0 / \rho$, which diverges to $-\infty$ as $d \rightarrow \infty$.*

Proof. Step 1: Compute the NPV differential.

By Definition ??:

$$\begin{aligned} V_{\text{rev}} - V_{\text{cont}} &= \left(-S(d) + \frac{\Pi_0}{\rho} \right) - \frac{\Pi_{\text{AI}}(d)}{\rho} \\ &= -S(d) - \frac{\Pi_{\text{AI}}(d) - \Pi_0}{\rho}. \end{aligned}$$

Step 2: Bound each term.

From A2.1: $\Pi_{\text{AI}}(d) \geq (1 + \gamma)\Pi_0$, so $\Pi_{\text{AI}}(d) - \Pi_0 \geq \gamma\Pi_0$. Since $\gamma > 0$ (A2.1) and $\Pi_0 > 0$ (A2.2):

$$\frac{\Pi_{\text{AI}}(d) - \Pi_0}{\rho} \geq \frac{\gamma\Pi_0}{\rho} > 0.$$

From A4: $S(d) \geq c d^2$ with $c > 0$. Therefore:

$$V_{\text{rev}} - V_{\text{cont}} \leq -c d^2 - \frac{\gamma\Pi_0}{\rho}.$$

Step 3: Establish strict negativity.

Since $c > 0$, $d \geq 1$, $\gamma > 0$, $\Pi_0 > 0$, $\rho > 0$:

$$-c d^2 - \frac{\gamma\Pi_0}{\rho} \leq -c - \frac{\gamma\Pi_0}{\rho} < 0.$$

Therefore $V_{\text{rev}}(d, \rho) - V_{\text{cont}}(d, \rho) < 0$ for all $d \geq 1$ and $\rho > 0$. This proves claim (i).

Step 4: Claim (ii) — individual irreversibility.

Since $V_{\text{rev}} - V_{\text{cont}} < 0$, we have $V_{\text{cont}} > V_{\text{rev}}$ for every $d \geq 1$ and $\rho > 0$. No organization at depth $d \geq 1$ can improve its NPV by reverting to pre-AI. Reversion is never individually economically rational. Claim (ii) is established.

Step 5: Claim (iii) — strengthening with depth.

The upper bound $-c d^2 - \gamma\Pi_0/\rho$ is a strictly decreasing function of d (since $-c d^2$ is strictly decreasing for $c > 0$, $d > 0$). As $d \rightarrow \infty$, this bound diverges to $-\infty$, so the NPV of reversion becomes arbitrarily negative. Claim (iii) is established.

Conclusion. For every organization at AI integration depth $d \geq 1$ and every positive discount rate $\rho > 0$, reversion to pre-AI is never economically rational: the NPV differential is bounded above by $-c d^2 - \gamma\Pi_0/\rho < 0$, and this bound tightens with increasing depth. AI adoption is individually economically irreversible at every integration depth $d \geq 1$. $\square$